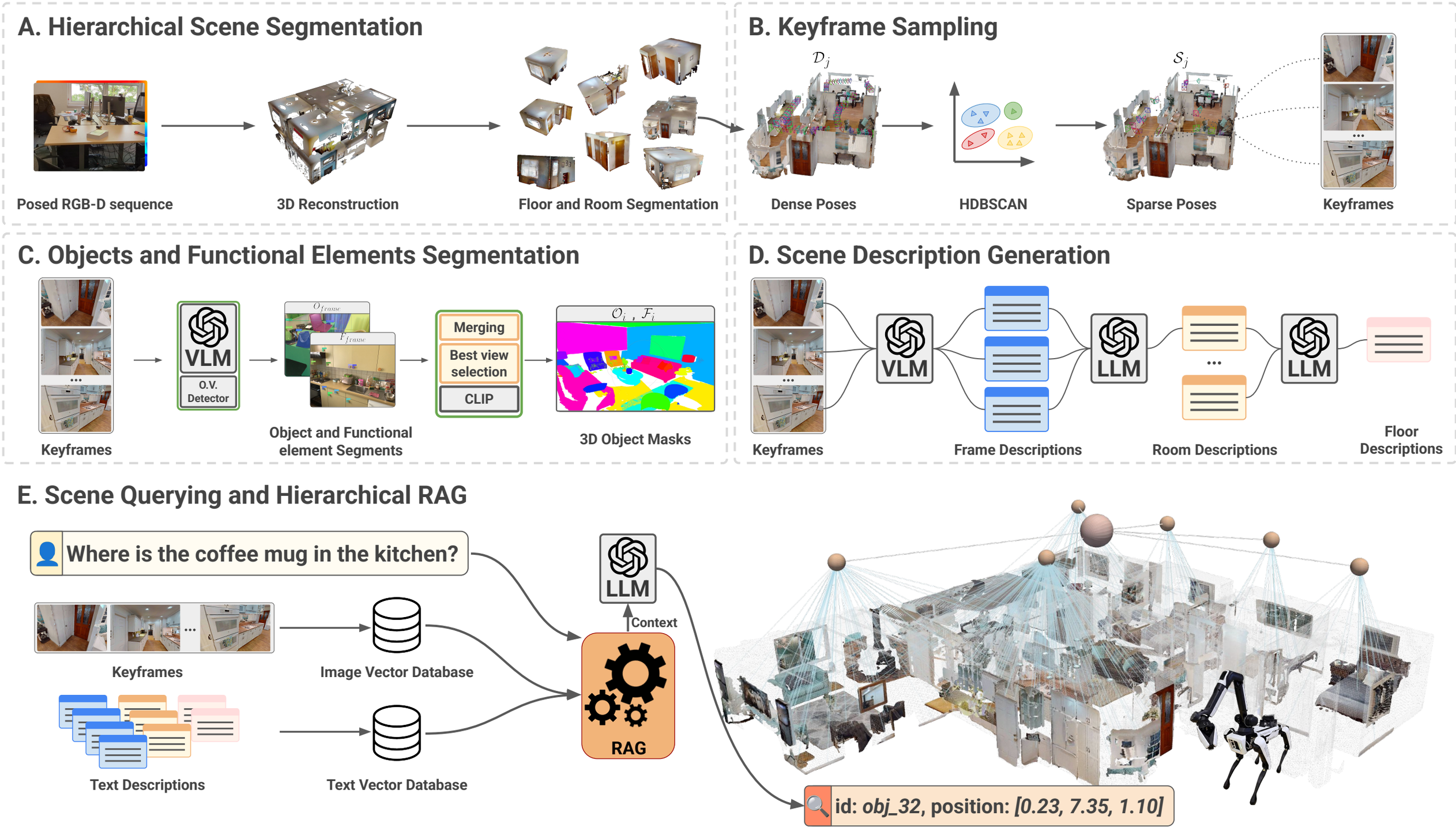
A. Hierarchical Scene Segmentation
Posed RGB-D sequence
3D Reconstruction
Floor and Room Segmentation
B. Keyframe Sampling
$\mathcal{D}_j$
$\mathcal{S}_j$
Dense Poses
HDBSCAN
Sparse Poses
Keyframes
C. Objects and Functional Elements Segmentation
VLM
O.V. Detector
O_{frame}
F_{frame}
Merging
Best view selection
CLIP
$\mathcal{O}_i$, $\mathcal{F}_i$
Keyframes
Object and Functional element Segments
3D Object Masks
D. Scene Description Generation
VLM
LLM
LLM
Keyframes
Frame Descriptions
Room Descriptions
Floor Descriptions
E. Scene Querying and Hierarchical RAG
Where is the coffee mug in the kitchen?
LLM
Context
RAG
Keyframes
Image Vector Database
Text Descriptions
Text Vector Database
id: obj_32, position: [0.23, 7.35, 1.10]